144 / Board tile: Top left

Cut around the outer grid boundary. Tape the four tiles together on the back. Each square is 25 x 25 mm; the assembled board is 300 x 300 mm.

Join right edge to top right; bottom edge to bottom left.

A1	B1	C1	D1	E1	F1
A2	B2	C2	D2	E2	F2
A3	B3	C3	D3	E3	F3
A4	B4	C4	D4	E4	F4
A5	B5	C5	D5	E5	F5
A6	B6	C6	D6	E6	F6

This line must measure 100 mm.

144 / Board tile: Top right

Cut around the outer grid boundary. Tape the four tiles together on the back. Each square is 25 x 25 mm; the assembled board is 300 x 300 mm.

Join left edge to top left; bottom edge to bottom right.

G1	H1	I1	J1	K1	L1
G2	H2	I2	J2	K2	L2
G3	H3	I3	J3	K3	L3
G4	H4	I4	J4	K4	L4
G5	H5	I5	J5	K5	L5
G6	H6	I6	J6	K6	L6

This line must measure 100 mm.

144 / Board tile: Bottom left

Cut around the outer grid boundary. Tape the four tiles together on the back. Each square is 25 x 25 mm; the assembled board is 300 x 300 mm.

Join top edge to top left; right edge to bottom right.

A7	B7	C7	D7	E7	F7
A8	B8	C8	D8	E8	F8
A9	B9	C9	D9	E9	F9
A10	B10	C10	D10	E10	F10
A11	B11	C11	D11	E11	F11
A12	B12	C12	D12	E12	F12

This line must measure 100 mm.

144 / Board tile: Bottom right

Cut around the outer grid boundary. Tape the four tiles together on the back. Each square is 25 x 25 mm; the assembled board is 300 x 300 mm.

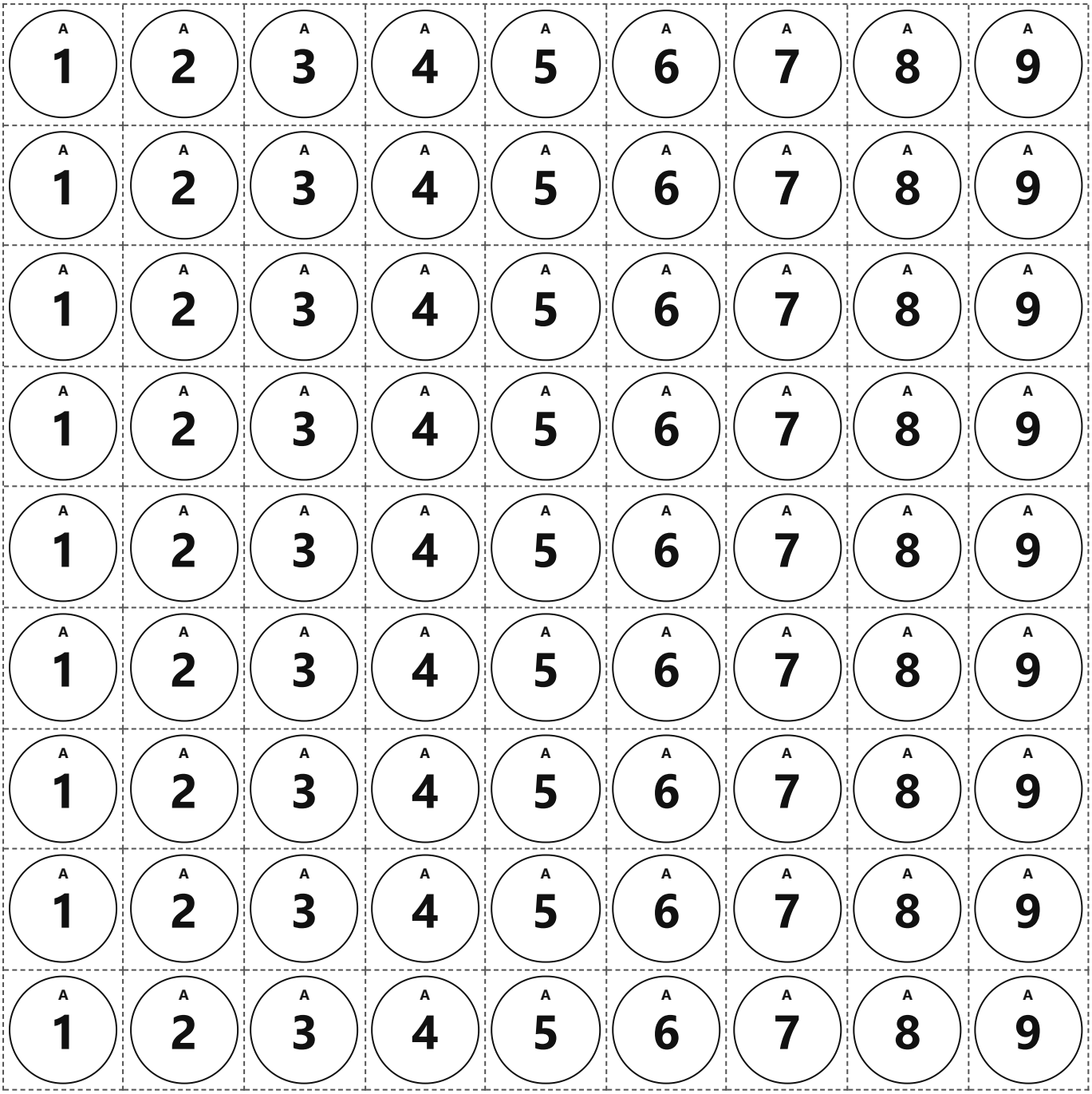
Join top edge to top right; left edge to bottom left.

G7	H7	I7	J7	K7	L7
G8	H8	I8	J8	K8	L8
G9	H9	I9	J9	K9	L9
G10	H10	I10	J10	K10	L10
G11	H11	I11	J11	K11	L11
G12	H12	I12	J12	K12	L12

This line must measure 100 mm.

144 / Player A: 81 pieces

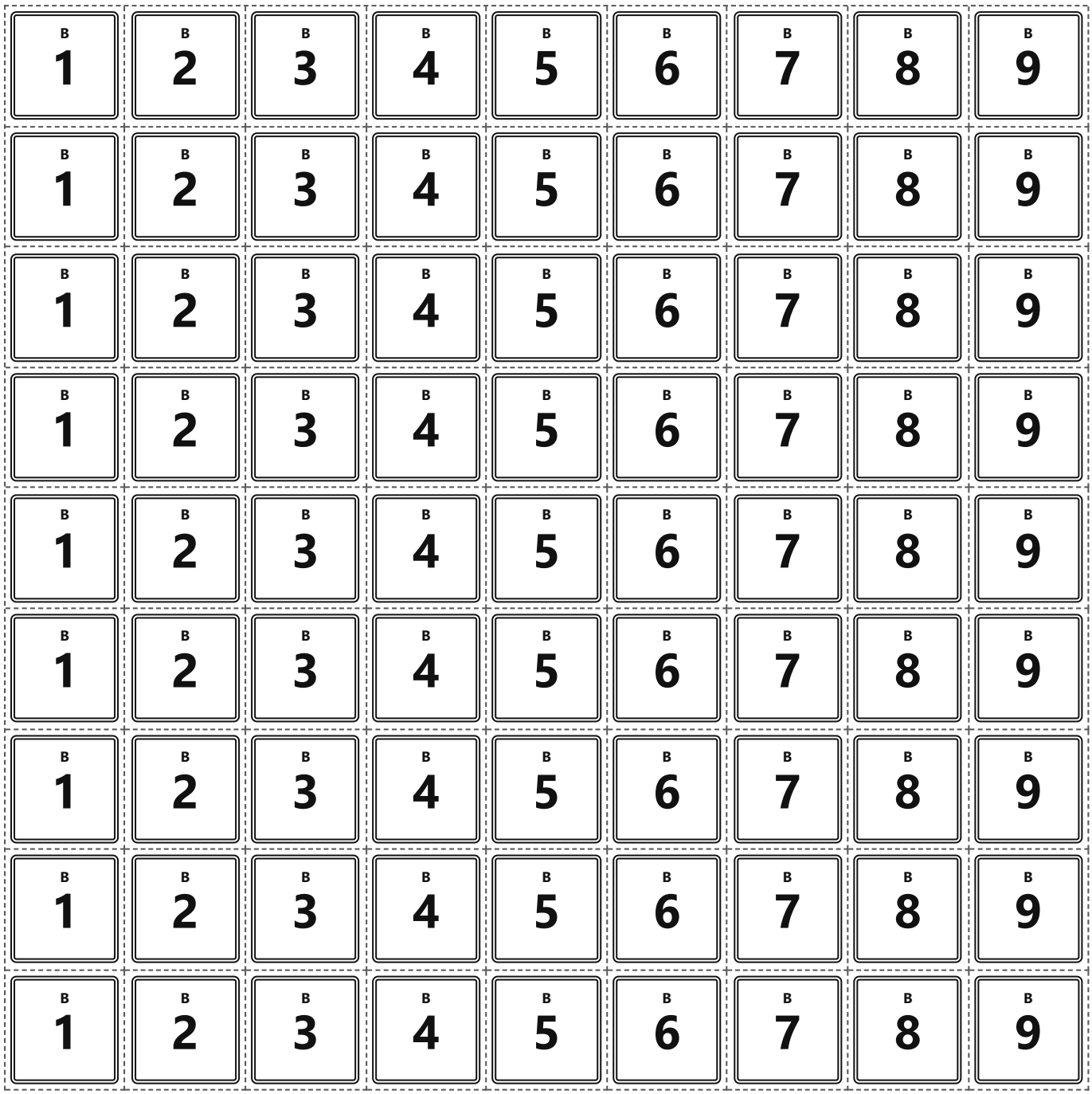
Nine copies each of 1 through 9. Cut the dashed 20 mm squares, then optionally trim A's circles. Player A has circular single-border marks; B has square double-border marks. Both sets work in black and white. Glue to card if desired.



This line must measure 100 mm.

144 / Player B: 81 pieces

Nine copies each of 1 through 9. Cut the dashed 20 mm squares, then optionally trim A's circles. Player A has circular single-border marks; B has square double-border marks. Both sets work in black and white. Glue to card if desired.



This line must measure 100 mm.

144 / Shared rule sheet

Confirmed game rules

Two players share a 12 by 12 board (144 squares) and 162 numbered pieces. Alternate turns: place a number from your reserve, or move one of your own pieces one square. First to make an adjacent ordered **3-6-9, 1-5-7, or 2-4-8** sequence of their own pieces wins. The three squares must be in a straight horizontal, vertical, or diagonal line.

Prototype choices, not additional user-confirmed rules

Player A starts. Each player has 81 pieces: nine copies of each digit 1 through 9. A uses circle pieces; B uses square pieces. Start with an empty board and keep unused pieces off the board as your reserve.

On a turn, do exactly one action: place one reserve piece on any empty square, or move your piece to an empty adjacent square (including diagonals). Moving does not consume a reserve piece. There is no capture, swapping, overwriting, skipping a turn, or movement beyond one square.

Read winning sequences in either direction. For example, 9-6-3 also wins when read backwards. No gaps, corners, board-edge wrapping, or mixed ownership. 1-7-5 is not 1-5-7. Check for a win immediately after each action; if it wins, stop before any draw check.

A draw occurs if the next player has no legal action (including a full board), the same board and player-to-move occur three times, or 200 consecutive moves occur without placing a piece. A placement resets that 200-move counter. There is also a 4096-turn prototype safety limit.

For paper play, keep a turn tally and a consecutive-move tally. Either player may request a position record before a suspected repetition: note occupied coordinates, owner and digit plus the next player. A disputed repetition without records is not a verified draw. You may agree to a draw at the table; that social agreement is outside the app's automatic rules.

The app lets you undo one completed turn at a time, including the final turn. Undo restores reserves, player-to-move and draw counters. For paper play, record each action (for example A: place 3 at C4; B: move D5 to E6) and reverse the last action only with both players' agreement.

Screen controls

Choose a reserve number, then touch an empty square. To move, select your own board piece, then a dashed adjacent square. Clear selection cancels without using a turn. Swipe or scroll to explore; square size never falls below 44 pixels. Keyboard: Tab into the board, arrow keys to navigate, Enter or Space to select or act. On phones the reserve stays at the bottom; swipe it sideways to reach every digit. Invalid actions do not change the turn. New game asks for confirmation. Saves belong to this device and browser; storage failures appear below the controls.

144 / Mathematical reference

What maps to gameplay

12 x 12 = 144 supplies the board size. 3 x 3 x 3 x 6 = 162 supplies the total inventory; the equal 81-piece sets and nine copies per digit are prototype choices. 369, 157 and 248 supply the three ordered winning families. The earlier 19 x 19 x 19 = 6859 voxel concept is separate.

Reference only: not game mechanics

Material	Values and relationship
Governors / anchor	PRIMARY_GOVERNOR 0.062483; SECONDARY_GOVERNOR 0.061573; rounded reciprocal MASTER_ANCHOR 16.0044.
Love scalars	11.0454 x 3.2413 x 1.7177 = 61.496159287854; catalogue rounds to 61.496159.
Signatures	248 and 157: difference 91, sum 405.
Keys	23297 and 97322 are digit permutations, not literal reversal. 23297 is prime; 97322 is composite. Difference 74025; sum 120619.
Relay	29, 23, 27, 37, 97, 39, 29 closes a six-node cycle.
Progression / macro	7 + 13 + 19 = 39; 384 = 6 x 4^3; 384 x 39 = 14976.
Earlier volume ladder	8, 144, 1728, 20736, 248832, 82944, 13824, 1536, origin (0 or 1), 162.

Decade families ending in 1, 5, 7 and 2, 4, 8 are selected residue classes, not a complete grid and not all prime numbers. The reference is the user's mathematical context, not established physics or evidence of game balance.

Full 45-value catalogue

0.000910, 0.0025, 0.061573, 0.062483, 1.7177, 2, 3, 3.2413, 6, 7, 8, 9, 10, 11.0454, 12, 13, 16.0044, 19, 23, 24, 27, 29, 37, 39, 61.496159, 81, 91, 97, 144, 157, 162, 248, 384, 405, 1536, 1728, 13824, 14976, 20736, 23297, 74025, 82944, 97322, 120619, 248832.